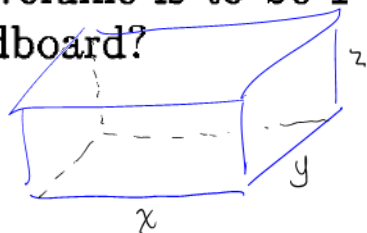


MA1521 L1 Week 06 Mon 20260914

left/right

front/back

A rectangular produce box is to be made of cardboard; the sides of single thickness, the ends of double thickness, and the bottom of triple thickness, with the top left open. Its volume is to be 1 cubic foot; what should be its proportions in order to use the least cardboard?



Volume is xyz , fixed as 1. ($\Rightarrow z = \frac{1}{xy}$)

$$A := \text{amt of cardboard} = 2 \times \underbrace{1 \cdot yz}_{\text{sides}} + 2 \times \underbrace{2 \cdot xz}_{\text{ends}} + \underbrace{3 \cdot xy}_{\text{btm}}$$

Want to minimize

$$A = \frac{2}{x} + \frac{4}{y} + 3xy$$

$$\text{Have } \left. \begin{aligned} A_x &= -\frac{2}{x^2} + 3y \\ A_y &= -\frac{4}{y^2} + 3x \end{aligned} \right\}$$

At min pt, have $A_x = 0 = A_y$

$$\Leftrightarrow \begin{cases} 3y = \frac{2}{x^2} \text{ and } 3x = \frac{4}{y^2} \end{cases} \quad (*)$$

[[Exploit symmetry!]]

$$\Leftrightarrow 3xy = \frac{2}{x} = \frac{4}{y}$$

$$\Rightarrow y = 2x$$

So at min pt, get (from $(*)$) $6x^3 = 2$, so $x = \frac{1}{\sqrt[3]{3}}$

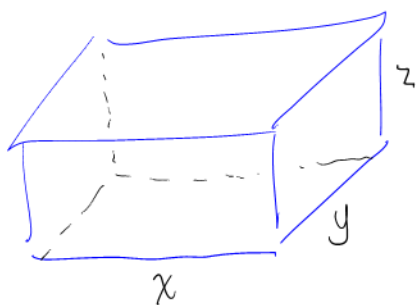
$$\text{so } y = \frac{2}{\sqrt[3]{3}} \text{ and } z = \frac{1}{\frac{2}{\sqrt[3]{3}}^2} = \frac{\sqrt[3]{3}}{2}$$

$$\text{so min pt } (x, y, z) = \frac{1}{\sqrt[3]{3}} \left(1, 2, \frac{3}{2} \right)$$

$$= \frac{3}{2\sqrt[3]{3}}$$

(Repeat of a previous question; but this time, use Lagrange multipliers.)

A rectangular produce box is to be made of cardboard; the sides of single thickness, the ends of double thickness, and the bottom of triple thickness, with the top left open. Its volume is to be 1 cubic foot; what should be its proportions in order to use the least cardboard?



$A :=$ amt of cardboard $= 2 \times \underbrace{1 \cdot yz}_{\text{sides}} + 2 \times \underbrace{2 \cdot xz}_{\text{ends}} + \underbrace{3 \cdot xy}_{\text{btm}}$
 $\uparrow$
 Want to minimize
 under constraint $V := xyz$ fixed as 1

Lagrange multiplier:

at minimum A , there exists $\lambda \in \mathbb{R}$

such that $\nabla A = \lambda \cdot \nabla V$ (& $V = 1$)

i.e. $\begin{pmatrix} 4z + 3y \\ 2z + 3x \\ 2y + 4x \end{pmatrix} = \lambda \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}$ & $xyz = 1$

Thus $\lambda = \frac{4z + 3y}{yz} = \frac{2z + 3x}{xz} = \frac{2y + 4x}{xy}$

i.e. $\lambda = \frac{4}{y} + \frac{3}{z} = \frac{2}{x} + \frac{3}{z} = \frac{2}{x} + \frac{4}{y} \Rightarrow \frac{2}{x} = \frac{4}{y} = \frac{3}{z} \Rightarrow \frac{\lambda}{2}$

$$\text{So } \underline{\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{2}{\lambda} \begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix}}$$

$$\text{Then } \boxed{xyz = 1} \Rightarrow \frac{192}{\lambda^3} = 1$$

$$\Rightarrow \underline{\lambda = \sqrt[3]{3} \cdot 4}$$

$$\text{So } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{\sqrt[3]{3}} \begin{pmatrix} 1 \\ 2 \\ 3/2 \end{pmatrix}$$

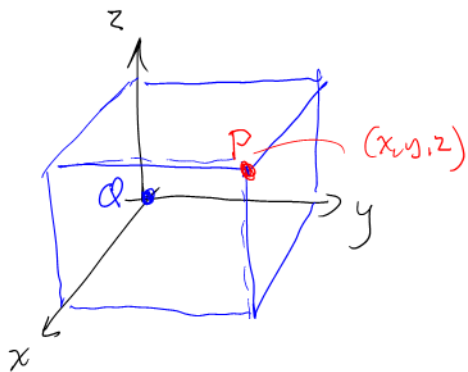
$$x > 0, y > 0, z > 0$$

A rectangular box is placed in the first octant so that one corner Q is at the origin and the three sides adjacent to Q lie in the coordinate planes. The corner P diagonally opposite Q lies on

a) the plane $x + 2y + 3z = 18$

b) the ellipsoid $x^2 + 2y^2 + 4z^2 = 12$.

Using Lagrange multipliers, tell for which point P the box will have the largest volume,
(and tell how you know it gives a maximum point.)



a) Want to maximize $f(x, y, z) = xyz$ (volume)
under constraint $\underbrace{x + 2y + 3z = 18}_{g(x, y, z)} \quad (x > 0, y > 0, z > 0)$

Lagrange multipliers : at max for f , have $\lambda \in \mathbb{R}$ s.t.

$$\boxed{\nabla f = \lambda \cdot \nabla g} \quad \left(\begin{array}{l} \& \boxed{x + 2y + 3z = 18} \\ \& x > 0, y > 0, z > 0 \end{array} \right)$$

i.e. $\boxed{\begin{pmatrix} yz \\ xz \\ xy \end{pmatrix} = \lambda \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}}$

i.e. $\lambda = yz = \frac{xz}{2} = \frac{xy}{3}$

$$\text{i.e. } y = \frac{x}{2}, \quad \frac{z}{2} = \frac{y}{3}, \quad z = \frac{x}{3}$$

$$\underline{x = 2y}$$

$$\underline{x = 3z}$$

$$\text{so } \underline{\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3z \\ 3z/2 \\ z \end{pmatrix} = z \begin{pmatrix} 3 \\ 3/2 \\ 1 \end{pmatrix}}$$

Then $\boxed{x + 2y + 3z = 18}$ gives $3z + 3z + 3z = 18$, so $z = 2$

i.e. $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ 2 \end{pmatrix}$ is coord for P

$$y = 3$$

$$x = 6$$

b) Want to maximize $f(x, y, z) = xyz$ (volume)

under constraint $\underbrace{x^2 + 2y^2 + 4z^2}_{g(x, y, z)} = 12 \quad (x > 0, y > 0, z > 0)$

Lagrange multipliers : at max for f , have $\lambda \in \mathbb{R}$ s.t.

$$\boxed{\nabla f = \lambda \cdot \nabla g} \quad \left(\begin{array}{l} \& \boxed{x^2 + 2y^2 + 4z^2 = 12} \\ \& x > 0, y > 0, z > 0 \end{array} \right)$$

$$\text{i.e. } \boxed{\begin{pmatrix} yz \\ xz \\ xy \end{pmatrix} = \lambda \cdot \begin{pmatrix} 2x \\ 4y \\ 8z \end{pmatrix}}$$

$$\text{i.e. } \begin{array}{lll} yz = \lambda 2x & \longrightarrow & x \cdot x \\ xz = \lambda 4y & \longrightarrow & x \cdot y \\ xy = \lambda 8z & \longrightarrow & x \cdot z \end{array}$$

$$\text{get } xyz = 2\lambda x^2 = 4\lambda y^2 = 8\lambda z^2 \quad \text{--- } \textcircled{*}$$

$$\text{so } x^2 = \frac{xyz}{2\lambda}, \quad 2y^2 = \frac{xyz}{2\lambda}, \quad 4z^2 = \frac{xyz}{2\lambda}$$

$$\text{so } \boxed{x^2 + 2y^2 + 4z^2 = 12} \quad \text{gives } 3 \cdot \frac{xyz}{2\lambda} = 12, \quad \text{so } \underline{xyz = 8\lambda}$$

$$\text{so } \textcircled{*} \text{ gives } 8 = 2x^2 = 4y^2 = 8z^2$$

$$\text{so } x > 0, y > 0, z > 0 \quad \text{gives } \begin{array}{l} x = \sqrt{4} = 2 \\ y = \sqrt{2} \\ z = \sqrt{1} = 1 \end{array} \quad \text{so } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ \sqrt{2} \\ 1 \end{pmatrix}$$

is where P is.

Using Lagrange multipliers, tell which point P in the first octant and on the surface $x^3 y^2 z = 6\sqrt{3}$ is closest to the origin.

Minimize $f(x, y, z) = x^2 + y^2 + z^2$ under constraint $g(x, y, z) = x^3 y^2 z$ fixed at $6\sqrt{3}$
 (& $x > 0, y > 0, z > 0$)

Lagrange multiplier: at minimum f , there exists $\lambda \in \mathbb{R}$

such that $\nabla f = \lambda \cdot \nabla g$ (& $g = 6\sqrt{3}$)

i.e. $\begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix} = \lambda \begin{pmatrix} 3x^2 y^2 z \\ 2x^3 y z \\ x^3 y^2 \end{pmatrix}$ & $x^3 y^2 z = 6\sqrt{3}$ & $x > 0, y > 0, z > 0$

so $\lambda = \frac{2}{3x y^2 z} = \frac{2}{2x^3 z} = \frac{2z}{x^3 y^2}$

$\Rightarrow 3y^2 = 2x^2$ and $2z^2 = y^2$ and $3z^2 = x^2$

Then $x^3 y^2 z = 6\sqrt{3}$ gives $6\sqrt{3} = (\sqrt{3} z)^3 (\sqrt{2} z)^2 z = 6\sqrt{3} z^6$, so $z^6 = 1$
 (& $x > 0, y > 0, z > 0$)

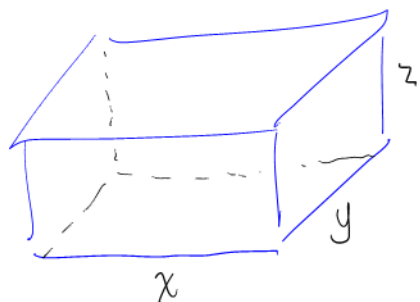
Then $x > 0$, $y > 0$, $z \geq 0$ gives $z = 1$, $y = \sqrt{2}$, $x = \sqrt{3}$

so point is $(\sqrt{3}, \sqrt{2}, 1)$, distance $\sqrt{3+2+1} = \sqrt{6}$ from origin.

In an open-top wooden drawer, the two sides and back cost \$2/sq.ft., the bottom \$1/sq.ft. and the front \$4/sq.ft. Using Lagrange multipliers, show that the following problems lead to the same set of three equations in λ , plus a different fourth equation, and they have the same solution.

a) Find the dimensions of the drawer with largest capacity that can be made for a total wood cost of \$72.

b) Find the dimensions of the most economical drawer having volume 24 cu. ft.



$$\text{Volume } V(x, y, z) := xyz$$

$$\begin{aligned} \text{Cost } C(x, y, z) &:= \underbrace{1 \cdot xy}_{\text{bottom}} + \underbrace{2 \cdot 2 \cdot yz}_{\text{two sides}} + \underbrace{2xz}_{\text{back}} + \underbrace{4xz}_{\text{front}} \\ &= xy + 4yz + 6xz \end{aligned}$$

For a), maximize V under constraint $C = 72$ (& $x, y, z \geq 0$)

Lagrange multipliers : at max for V , there exists $\lambda \in \mathbb{R}$ such that

$$\boxed{\nabla V = \lambda \cdot \nabla C} \quad \text{and} \quad \boxed{C = 72} \quad (\& \ x, y, z \geq 0)$$

$$\text{i.e. } \underbrace{\begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}}_{\text{red underline}} = \lambda \begin{pmatrix} y+6z \\ x+4z \\ 4y+6x \end{pmatrix} \quad \text{and} \quad \underline{xy + 4yz + 6xz = 72} \quad (\& \ x, y, z \geq 0)$$

For b), minimize C under constraint $V=24$ (& $x, y, z \geq 0$)

Lagrange multipliers : at min for C , there exists $\mu \in \mathbb{R}$ such that

$$\boxed{\nabla C = \mu \cdot \nabla V} \quad \text{and} \quad \boxed{V=24} \quad (\& x, y, z \geq 0)$$
$$\text{i.e. } \underline{\begin{pmatrix} y+6z \\ x+4z \\ 4y+6x \end{pmatrix} = \mu \begin{pmatrix} yz \\ xz \\ xy \end{pmatrix}} \quad \text{and} \quad \underline{xyz = 24} \quad (\& x, y, z \geq 0)$$

The Lagrange multipliers equations for a) and b) are equivalent :

simply take $\mu = \frac{1}{\lambda}$

Soln for a) : $\underline{\begin{pmatrix} yz \\ xz \\ xy \end{pmatrix} = \lambda \begin{pmatrix} y+6z \\ x+4z \\ 4y+6x \end{pmatrix}} \Rightarrow \frac{xyz}{\lambda} = xy+6xz = xy+4yz = 4yz+6xz$

$$\Rightarrow \underline{3x=2y \text{ and } y=6z \text{ and } x=4z}$$

Then $\underline{xy+4yz+6xz=72}$ gives $72 = (4z)(6z) + 4(6z)z + 6(4z)z$

$$= \underline{72z^2}$$

Since $x, y, z \geq 0$, we get $z=1$, $y=6$, $x=4$, so $(x, y, z) = (4, 6, 1)$

* Note : The resulting max V value here is $4 \cdot 6 \cdot 1 = 24$ (same as constraint for b))

Soln for b)

$$\begin{aligned} \left(\begin{array}{c} y+6z \\ x+4z \\ 4y+6x \end{array} \right) &= \mu \left(\begin{array}{c} yz \\ xz \\ xy \end{array} \right) \Rightarrow \mu xyz = xy+6xz = xy+4yz = 4yz+6xz \\ &\Rightarrow \underline{3x=2y \text{ and } y=6z \text{ and } x=4z} \\ &\hspace{15em} (\text{as before}) \end{aligned}$$

Then $xyz = 24$ gives $24 = (4z)(6z)z = 24z$

Since $x, y, z \geq 0$, we get $z=1$, $y=6$, $x=4$, so $(x, y, z) = (4, 6, 1)$
(as before)

* Note: The resulting min C value here is $(4)(6) + 4(6)(1) + 6(4)(1) = 72$
(same as constraint for a)